Sungsu Jeon

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RESEARCH INTEREST

Electrochemical Engineering, Sustainable Energy Conversion, Critical Metals Separation

EDUCATION

Yonsei University, B.S. in Chemical and Biomolecular Engineering — Mar. 2021 – Feb. 2027 (Expected)
- GPA: 4.05 / 4.3

Purdue University, Exchange Program in Chemical Engineering — Aug. 2025 – May. 2026
- GPA: 3.97 / 4.0

RESEARCH EXPERIENCE

Dong Research Group, Purdue University — Aug. 2025 – Jul. 2026

Undergraduate Research Assistant, Advisor: Prof. Qi (Tony) Dong

Project Title: Programmable Non-Equilibrium Electrodeposition for Sustainable Cobalt–Nickel Separation

- Developed a programmable pulsed electrodeposition system for Co/Ni separation, integrating cyclic voltammetry and machine learning-guided optimization to achieve 91% Co selectivity.
- Investigated Co/Ni deposition and stripping kinetics using electrochemical techniques (CP, EQCM, RDE, and Tafel analysis), with separation selectivity quantified by ICP-OES.
- Evaluated scale-up feasibility by examining electrode area and electrolyte concentration effects, and modeled a multistep separation process predicting 98% Co purity from Co-rich feeds.

Energy Conversion and Storage Laboratory, Yonsei University — Sep. 2024 – Aug. 2025

Undergraduate Researcher, Advisor: Prof. Jong Hyeok Park

B.S. Thesis: Surface Reconstruction and C_{2+} Selectivity of CuO Nanosheets in Acidic Electrochemical CO_2RR

- Synthesized CuO nanosheets using hydrothermal method and fabricated gas diffusion electrodes via spray coating.
- Evaluated acidic CO_2 electroreduction performance in flow-cell system, achieving 80% C_{2+} Faradaic efficiency at 100 mA cm^{-2}, with products quantified by GC and ^{1}H NMR.
- Identified gas loss through the catholyte outlet as the cause of apparent Faradaic efficiency deficits at high current densities by comparing PTFE and carbon-paper GDLs.
- Observed CuO nanosheet reconstruction into nanoparticle aggregates after electrolysis using SEM.

Review Project in Integrated Science and Engineering, Yonsei University — Jul. 2021 – Feb. 2022

Research Intern, Advisor: Prof. Rajkumar Patel

Project Title: Membrane Technologies for Pressure Retarded Osmosis (PRO) and Recovery of Valuable Minerals

- Conducted a review of over 50 peer-reviewed publications on membrane technologies for pressure-retarded osmosis (PRO) and mineral recovery from seawater, resulting in the publication of two first-author review papers.

PUBLICATIONS AND MANUSCRIPTS

[4] Priyam Mondal[†], **Sungsu Jeon**[†], Chenlu Yang[†], Ashutosh Rana, Chase Presz, Su Youn Kim, James Chen, James Nguyen, Cheng Zhu, Udayan Singh, Jihye Kim, Shuwen Yue*, Jeffrey E. Dick*, and Qi Dong*, "Non-equilibrium Electrochemical Choreography for Sustainable Cobalt-Nickel Separation," *submitted* ([†]Equal Contribution)

[3] Priyam Mondal, Yuting Xu, Yao Shan, Adam John Smerigan, Michael Schwebach, **Sungsu Jeon**, Wenqin Li, Fanglin Che*, Qi Dong*, "Programmable, Ultrafast, and High-temperature Extraction of Critical Metals from Spent Lithium-ion Batteries," *in preparation*

[2] **Sungsu Jeon** and Rajkumar Patel*, "Recovery of Valuable Minerals from Sea Water by Membrane Separation and Adsorption Process: A Review," *Membrane Journal*, vol. 32, no. 1, pp. 13–22, 2022. https://doi.org/10.14579/MEMBRANE_JOURNAL.2022.32.1.13

[1] **Sungsu Jeon**, Rajkumar Patel, and Jonghak Kim*, "Review on the Recent Membrane Technologies for Pressure Retarded Osmosis," *Membrane Journal*, vol. 31, no. 4, pp. 253–261, 2021. https://doi.org/10.14579/MEMBRANE_JOURNAL.2021.31.4.253

PRESENTATIONS

[2] (Poster) **Sungsu Jeon**, Priyam Mondal, and Qi Dong*, "Programmable Non-Equilibrium Pulsed Electrodeposition for Sustainable Cobalt-Nickel Separation," *Spring Undergraduate Research Conference*, *Purdue University*, West Lafayette, IN, Apr. 2026.

[1] (Poster) Zixi Fan, **Sungsu Jeon**, Michael Yackovich, Hanzhe Liu, and Qi Dong*, "Structure-Function Relationship in Cu-cellulose MOF for Ion Transport and Electrocatalysis," *10th Notre Dame–Purdue Symposium on Soft Matter & Polymers*, West Lafayette, IN, Oct. 2025.

RESEARCH SKILLS

Electrochemical Techniques: CV/LSV, Chronoamperometry, Chronopotentiometry, Electrochemical Mass Spectrometry (EC-MS), Electrochemical Quartz Crystal Microbalance (EQCM), Rotating Disk Electrode (RDE)

Reactor & Electrode Fabrication: Flow Cell, H-Cell, Gas Diffusion Electrode (GDE) Spray Coating, Drop Casting, Gas Diffusion Layer (GDL) Handling (Sigracet, PTFE)

Characterization: ICP-OES, GC, ^{1}H NMR, SEM, XRD

Material Synthesis: Hydrothermal Synthesis, Electrodeposition

Software & Data Analysis: BioLogic EC-Lab, CHI Workstation, Aspen Plus, Polymath, Python, MATLAB

HONORS & AWARDS

Honors

- **President's Award**, Korea Institute for Advancement of Technology (KIAT) — Nov. 2026
- **Dean's List & Semester Honors**, Purdue University — Fall 2025, Spring 2026
- **Honors** (top 10% of department), Yonsei University — Fall 2021, Fall 2024

Scholarships

- **Jinri Merit-Based Scholarship**, Yonsei University — Feb. 2022, Jul. 2024, Feb. 2025
- **Korea-U.S. Student Exchange Program** ($18,000), Ministry of Trade, Industry and Energy — Aug. 2025

TEACHING & SERVICE

Career Mentoring, Naver Blog — Dec. 2020 – Present

- Provided academic and career mentoring for high school and undergraduate STEM students.

Study Mentoring, KT Online Night School — Jul. 2024 – Dec. 2024

- Taught Math and English to underserved middle school students while supporting career exploration.

Sergeant, Republic of Korea Army — Jan. 2023 – Jul. 2024

- Served as a medic responsible for emergency response and medical support near the DMZ.